

Job Applications Selection and Identification: Study of Resumes with Natural Language Processing and Machine Learning

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Abstract—For each job ad, the organization received a high number of applicants. Finding the right person's application from various applications is a time-consuming task for any company these days. The categorization of a representative's resume is a time-consuming, labor-intensive, and resource-intensive procedure. Natural language processing and advanced analytics can understand and analyze unstructured written material to extract the required information. The idea is to teach the machine to study text in the same manner that people do. By reviewing applications with natural language processing and machine-learning processing, we may be able to locate people with the abilities and attributes we need more quickly. This study provides a comprehensive assessment of resume selection and identifies the existing work gaps. Different algorithms for machine learning and methodologies for analyzing and interpreting unstructured data have been researched. This study also addresses the research problems and future potential of resume analysis regarding the writing style, word choice, and grammar of unstructured written communication.

Keywords—Resume parser, Unstructured written language, Text Mining, Natural Language Processing, Machine Learning

I. INTRODUCTION

The resume is an official and professional document primarily used to demonstrate a job applicant's brief profile. A resume comprises information on a job applicant's education, abilities, experience, accomplishments, and portfolio. The corporate recruiting process is rapidly expanding in today's digitally advanced world. Hard copies of resumes are no longer accepted, and recruiting teams prefer a resume which can also be viewed online. In the current system, the prospect must fill out each detail of their Job application manually, which takes a long time, and the applicants are displeased with the job that the current system chooses based on their qualifications. Based on their qualifications, expertise, and other information on their profile, resume screening and evaluation assess if an applicant is qualified for a position. It involves comparing resume-based qualifications of candidates to job criteria like pattern matching. In global terms, each professional segment has evolved its standards for actionable insights into each candidate's capabilities. As there is no universally accepted

format for a resume, some candidates feel that presenting only basic relevant information is more than sufficient, while others aim to promote themselves as multitasking persons. Nevertheless, others believe the personal interview is essential, while the resume is a minor issue. The practice of collecting information from webpages or unstructured texts like documents or PDFs using sophisticated pattern recognition or language analysis tools is known as resume analysis. It is a method of mechanically extracting information from resumes or unstructured papers and storing it in a database for recruiters. The human resource role in firms has developed over time, much as technology has been appreciated for its greater efficiency in practically every industry. So, for human resource management systems and recruiting professionals, extracting meaningful information from the resume is typically time-consuming.

II. RELATED WORKS

Currently, the HR department is in charge of the recruiting process. Human skills are frequently tested during recruiting exercises, such as obtaining, choosing, and analyzing resumes, contacting applications, offering feedback to successful and failed candidates, and conducting face-to-face interviews with various prospects. Among other things, time limits, prejudice, and bias are common human limitations that tend to hinder the efficacy and efficiency of recruiting activities. These constraints hinder hiring the best individuals for firm openings and can result in considerable monetary losses. In Abdul Wahab et al. [1] case, multiple NLP approaches such as bigram trigram and n-gram and text classification were utilized to choose the best candidates from a pool to accomplish these sorts of jobs; this model employs Machine Learning to accomplish the categorization using the methodology. The benefit is that the text corpus is converted into a list of words. Based on the concordance, allocate the word forms to separate lemmas. Natural Language Processing was employed in Tejaswini K et al. [16] research to extract information from an unstructured resume. It generates a condensed version of each résumé, containing just the elements relevant to the selection process: resume classification and indexing of resume similarities

with the cosine similarity method used to compute the similarity index. The similarity index indicates if the candidate is a good fit for the job. Using the SVM classifier, keywords from the application and interpretation offered an accuracy of around 78%. Existing resume analysers use

semantic search to understand the language context to provide reliable and complete results. Table 1 summarizes the various methods and algorithms used in the existing literature and the gaps in the literature.

TABLE 1: GAP IDENTIFIED OF THE EXISTING METHOD & ALGORITHM

Author	Year	Method & Algorithm	Gap Identified
Riya P [14]	2022	Naïve Bayes, Support Vector, Random Forest	Video interviewing features such as facial recognition, voice-to-text Generation and voice analysis
Lokesh. S [6]	2022	Logistic Regression, Support Vector, Decision Tree	Social networking sites not utilized
Tejaswini K [16]	2022	TF-IDF, cosine similarity, KNN	Content-Based Filtering Systems on the academic score and test results
Rabih H [13]	2021	Matcher Algorithm, Naïve Bayes, Support Vector, Random Forest	Decision and results obtained with French Higher Education systems
Pradeep R [8]	2020	Ranking using cosine similarity and k-NN, SVM	Poor accuracy
Mhamdi D [7]	2020	Cluster analysis	Word2vec method and self-organized learning
Suhas T [15]	2019	Naïve Bayes, Support Vector, Logistic Regression & NLPP module	Domain dependent
Jie C [5]	2018	Two-Step Information Extraction Algorithm	Multiple label predication, low recall rate of name entity classification
Yasunobu K [19]	2017	Job-related positive and negative keyword matching, Co-occurrence Network	Results depend on limited keywords
Tim Z [17]	2016	Filtering, ranking score, and comparison prototype	Weights to calculate scores are not optimized
Çelik D [4]	2013	Ontology-based resume parsing	Results depend on limited features

III. INFORMATION EXTRACTION METHODS

A. Resume Parsers

Natural language processing and machine learning can comprehend and analyze unstructured written material to extract the desired information. Existing resume parsers employ semantic search to comprehend the linguistic context and produce reliable and complete results. [3] A resume parser transforms unstructured data into structured data. Based on the parameters specified, the resume analyzer automatically splits the material into several areas. Separate data into fields depending on predefined criteria and factors, such as Name, address, email, start/end date, job title, company, city, title, university, previous year, location, dancing, singing, and swimming. For information extraction, various commercial and open-source resume parsers are available.

Resume screening technology automates manual labour and accelerates the selection process for countless people who have applied to various available positions within a firm. It eliminates the possibility of human mistakes, which is especially important given the enormous of candidates and the tight hiring deadlines. Overall, it assists recruiters in obtaining the finest applicants through a simplified, bias-free process. A resume parser examines and extracts text from various file types, including pdf, and document files, in real-time and ignores everything else. Resumes using tables, partitions, and

so on need higher-order intelligence from the programme. These resumes are categorized as complicated and are examined using Optical Character Recognition (OCR) combined with sophisticated Natural Language Processing (NLP) techniques. The retrieved text is keyword-classified and kept separately for each candidate. A resume is a collection of information about a candidate that may be maintained in many categories, such as Experience, Personal information, abilities, education, etc. Resume parsing in recruiting pulls information from resumes based on tags such as job title, business name, years of experience, university, and so on. However, this technique becomes more complicated when the number of competitors increases since the same firm's or universities' names may be written differently. Named Entity Recognition (NER) techniques are used in these cases.

B. Machine Learning

Machine learning is a technology used to do tasks that demand high intelligence. Machine learning is distinguished from other types of software by three key components: high-speed computation, a significant amount of high quality data, and complex algorithms [10-11]. By utilizing an algorithm that links excellent data, core machine learning technologies improve the accuracy and reliability of standard procedures.

TABLE 2 OPEN-SOURCE RESUME PARSERS

Resume Parser	Focuses on	Programming language	Library used	Output file	Advantage	Disadvantage
Brendan Herger's (Herger, 2015/2020)	Extracts information from resume	Python	PDFMiner	CSV file	Simple and Language dependent	Information loss in terms of date and job description
Skript Technologies' (Skript/Cvscan, 2016/2020)	Extracts information from resume	Python	PDFMiner	.json format	Simple and CLI interface	Fails to extract date most of the time
Antony Deepak's (GitHub—Antonydeepak/resume parser: resume parser using rule based approach. developed using framework provided by GATE, n.d.)	Extracts information from resume uses a hybrid machine-learning and rule-based approach focuses on semantic rather than syntactic parsing	Java	Apache's Tika library	Structured.json format	Better accuracy	Complex and difficult to use
Keras-English-resume-parser-and-analyzer (Chen 2018/2020)	Extracts information from English resume use Keras and deep learning models	Python	PDF miner	Raw content	Simple and better accuracy	Language dependent

C. Natural Language Processing and its role in resume Parsing

Natural language, such as voice or writing, may not necessarily adhere to rigid grammatical rules. As a result, the interpretation of those texts cannot rely simply on some fixed rules and setups. Because of cultural and linguistic developments, the spoken language varies from place to location. Even though English is spoken in the United States and the United Kingdom, the format and structure differ. If we train a model using data from US text and run it on text

from those in the UK, the outcomes will be skewed. Different sorts of text adhere to various formats. For example, menu cards at restaurants, wedding invites, and business emails have their formats, but resumes do not have any standard format. NLP examines the context of a text, not simply the words in it, by analyzing whether words are used frequently, negative or positive phrases, memorable characters, and other factors. Because resumes are much more organized than emails or text annotation, and we know which information points we anticipate uncovering ahead of time, utilizing NLP to analyze resumes can be considerably more straightforward and yield reasonably accurate results [9, 12].



Fig. 1. Keyword cloud resume screening [2]



Fig. 2. Trending jobs skills keyword cloud [18]

Figs. 1 and 2 shows the keyword cloud for the resume screening and trending job skills required in the candidate as per our survey. Fig 3 illustrates the general workflow of

the resume screening process while Fig 4 suggests the ranking process methodologies.

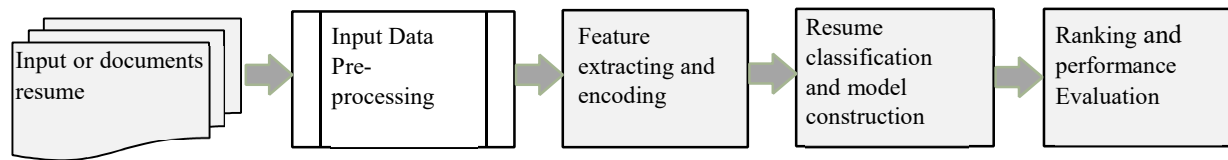


Fig. 3. General Structure of Resume Analysis in the literature

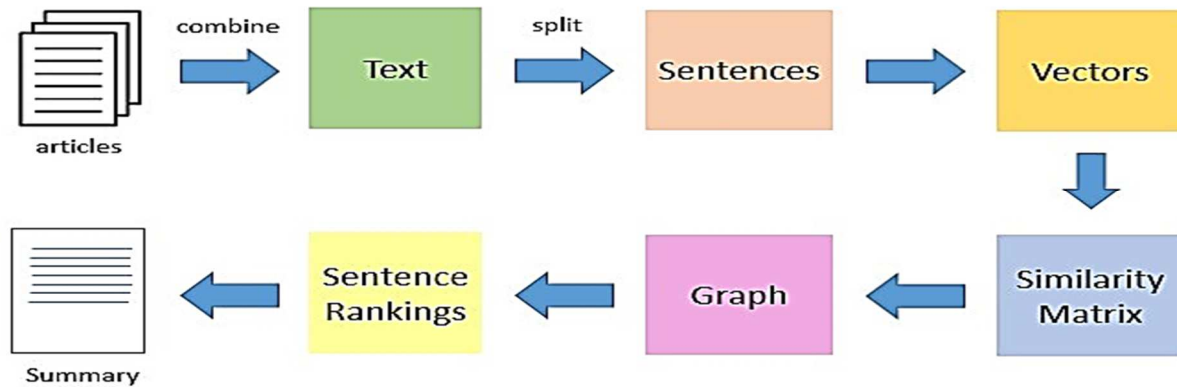


Fig. 4. Resume ranking Structure available in the literature

1. **Extract texts from files:** Real-life database may contain pdf, documents, images, and rich text formats.
2. **Pre-processing:** we need to clean out all the unnecessary things, such as symbols, punctuation marks, and non-informative words like pronouns, adverbs, and prepositions, with the help of regular expressions.
3. **Feature Extraction and Label Encoding:** Named Entity Recognition, Count, TF-IDF, Semantic, and Doc2vec features needs to be extracted.
4. **Resume Classification Model Construction:** Various machine learning algorithms were used in the literature at this stage, like Naïve Bayes, Linear Regression, Support Vector Machine, Random Forest, Decision Tree, and KNN.
5. **Ranking:** First, review resumes based on the minimal credentials for the position. Next, review resumes depending on the preferred credentials for the position. Next, scan resumes depending on the criteria of applicants we wish to interview. The score can be given to each stage with the help of vectorization and co-sign similarity matrix, and finally, a ranking would be given to each resume.
6. **Performance Evaluation:** Precision, Recall, F1 and Accuracy are the widely used evaluation metrics.

IV. RESEARCH CHALLENGES

Various elements, including the writing style, word choice, and syntax of the written text, can determine the accuracy of the resume analyzer. A combination of statistical algorithms and complicated criteria is required to know and acquire relevant information from resumes correctly. Natural language processing and machine learning can comprehend and evaluate unstructured written language and context-based data. [3] Machine learning is developing as a tool to help organizations perform talent sourcing and recruiting more efficiently. Even though automation has the potential to improve the effectiveness and capacity of recruitment, there

are specific issues to consider. Machine learning requires vast, high-quality data sets to give benefits. As a result, the approach may not be as practical for organizations that examine fewer resumes or do not hire regularly. There are several methods to write the same information, such as Name, address, and date, and the following are some crucial problems that must be looked after.

1. Recognize the resume writing style
2. Understand resume wording. Understand the grammar of the unstructured written language
3. Contextual search
4. Recognize the structure and format of resumes
5. Recognize resume headers and footers
6. Separate resume into parts.
7. Recognize structural and visual information in PDF documents
8. Scanner speed.

V. CONCLUSIONS

Resume classification with machine learning technology will deliver a better and more efficient hiring option. This will introduce the possible candidate to the organization, and the applicant will be put in an organization that values their skill set and abilities. We want to make the resume-scoring process easier. The finest software is intended to function in tandem with the current recruiting stack, causing little disruption to the candidate's workflow. This approach may be more adaptable by including a wide range of resume screening options. Natural language processing and machine learning can comprehend and evaluate unstructured written language and context-based data. A thorough study of resume selection that compares acknowledged works and research to open source and commercial resume analyzers are presented. This reviews different commercial and open-source resume parsers for information extraction. The following section provides an overview of how to utilize semantic search for context based search. This study also discusses the research constraints and

future potential of resume analysis regarding writing style, word choice, and unstructured syntactic written language. Resumes in the form of videos can also be acceptable. We investigated the implemented systems for organizing unstructured text data in a specific industry, namely recruiting and resource management. It is feasible to divide resumes into separate categories using feature engineering rather than sophisticated algorithms.

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